Action-Conditioned Predictive Consistency as a Diagnostic for Gaussian-Noise Robustness in JEPA World Models

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Abstract

Latent predictive world models avoid pixel reconstruction, but pixel-free prediction is not a robustness definition for closed-loop control. We study this gap with action-conditioned predictive consistency (ACPC), a no-retraining diagnostic for JEPA-style control checkpoints. The diagnostic combines ACPC Tail Risk (ATR), which measures high-quantile clean/noisy rollout disagreement for the same state under the same action sequence, with Selective Margin Pass Rate (SMPR), which checks that task-grounded near-boundary distinctions remain separated. Across three independent LeWM training seeds and four control tasks, no-noise checkpoints are fragile under observation-only Gaussian noise, while input-side noise training yields broad matched-Gaussian robustness plateaus. At recovered endpoints, ATR drops sharply and SMPR rises to near one across tasks. These results support ACPC as a bounded diagnostic for matched-Gaussian robustness plateaus in fixed JEPA world-model checkpoints.

Keywords: visual world models; JEPA; action-conditioned predictive consistency; visual robustness; predictive-dynamics diagnostics.

Code and data release: `https://github.com/Anguo-star/lewm-acpc-diagnostics`

1 Introduction

JEPAs predict future representations instead of reconstructing pixels, which can reduce pressure to encode high-magnitude irrelevant visual detail [1, 3, 29]. For control, however, "irrelevant" is action- and transition-dependent. A representation may ignore a background texture but must preserve a contact pose, doorway state, or object-goal relation if that distinction changes the next useful action. Encoder-level invariance is therefore an incomplete robustness target: planning uses the model's *predictions*, so the robustness question should be asked after action-conditioned rollout.

We study this gap as a controlled diagnostic problem for JEPA latent world-model control. Let a clean history and a visually perturbed history describe the same underlying state, and evaluate both branches under the same action sequence. The desired behavior is not necessarily identical encoder latents. It is that task-relevant predicted futures agree after the shared action intervention, while states that differ in action, transition, cost, or goal relation remain separable. This selective requirement is the core of action-conditioned predictive consistency (ACPC).

The selective tension matters because consistency alone can be misleading. A collapsed model can make clean and corrupted views agree while erasing distinctions needed for planning. Conversely, a large encoder shift can be harmless if the action-conditioned predictor contracts nuisance directions before they affect the predicted rollout used by planning. The diagnostic must therefore pair a same-state predictive-stability metric with an anti-collapse margin check for task-grounded near-boundary separations.

We instantiate this diagnostic with a fixed Gaussian intervention. Across PushT, TwoRoom, Reacher, and Cube, we evaluate no-noise LeWM [19] checkpoints and full-sequence input-side Gaussian-noise checkpoints over $\sigma_{\max} \in \{0.01, \dots, 0.08\}$. The primary endpoint is observation-only Gaussian evaluation noise with the goal image kept clean. Across three independent LeWM training seeds, the no-noise baseline remains strong on clean observations but loses 74.56 ± 5.55 points on PushT and 41.00 ± 1.44 points on Reacher at evaluation noise

$\sigma = 0.08$. Test-time visual fragility is not new [15, 33, 12, 21]; our focus is what it reveals about fixed JEPA world-model checkpoints.

The evidence shows broad matched-Gaussian recovery plateaus rather than a stable universal point optimum. Input-side noise training recovers the observation-noise endpoint on TwoRoom, PushT, and Reacher and gives a weaker but positive Cube signal across training seeds. At recovered endpoints, ACPC Tail Risk (ATR) drops sharply and Selective Margin Pass Rate (SMPR) rises to near one across tasks. Bounded blur/resize checks delimit perturbation scope.

This paper makes three contributions. First, it formulates visual robustness for JEPA world-model control as selective action-conditioned predictive consistency rather than encoder invariance. Second, it instantiates this diagnostic with ATR for same-state predictive tail risk and SMPR for task-grounded anti-collapse margins. Third, it provides a fixed-checkpoint Gaussian robustness study across four tasks and three LeWM training seeds, with bounded unseen-stressor checks that delimit the perturbation scope.

2 Related Work

This diagnostic sits between latent-prediction representation learning and robust visual-control methods. The related work is organized around three questions: what latent prediction promises, where augmentation-based invariance is insufficient for control, and how control-relevant representation work treats downstream predictive consequences.

2.1 Latent prediction and JEPA world models

JEPA [16] predicts in latent space rather than reconstructing pixels; I-JEPA [1] and the V-JEPA family [3, 2, 20] extend this idea to images, video, and dense physical-world representations. LeWM [19], our baseline, stabilizes an end-to-end JEPA world model with SIGReg [6]. PLDM [25, 26], via `stable-worldmodel` [18], provides related latent-dynamics planning context. LeJEPA theory [14] studies latent-variable recovery and latent planning; our diagnostic asks whether learned visual world-model checkpoints give paired clean/noised views consistent action-conditioned predictions.

2.2 Invariance, augmentation, and visual robustness

Augmentation and joint-embedding losses impose invariances only when their training signal identifies the right nuisance factors [31, 27, 35]. Recent theory clarifies why latent prediction can help with nuisance or noisy features but also why it is not enough for control. Van Assel et al. [29] and Littwin et al. [17] analyze representation-level effects; they do not address closed-loop action-conditioned rollout, candidate-cost stability, or the need to keep action-distinct states separable. Seq-JEPA [8] addresses a related invariance–equivariance tension architecturally.

We do not train robust visual-control systems as competing baselines. DrQ/DrQ-v2 [15, 33], SODA/DMC-GB [12], DreamerV3 [10], TD-MPC2 [11], and ViGMO [21] define future method-comparison axes; noisy/video JEPA variants [22, 13, 23] are representation-level context. ACPC is complementary: it tests whether same-state visual perturbations become equivalent after a fixed action intervention while action-, transition-, or cost-distinct cases remain discriminable.

2.3 Control-relevant predictive representations

Bisimulation merges states with equivalent reward and transition behavior. DeepMDP [7], DBC [34], bisimulation-regularized MPC [24], and Bisim-JEPA [28] use this principle to learn control-relevant representations. Value-equivalent and value-aware models [9, 30] and VIBR [5] likewise emphasize downstream control consequences. Wang et al. [32] formalize action-conditioned video world modeling as a group action with identity, inverse, and composition-consistency metrics. ReOI [4] treats visual distractor robustness in visual MPC through observation interventions for world-model action-outcome prediction. These lines show that downstream predictive consequences are already central to control-representation work. Our diagnostic instantiates that view for paired clean/noised JEPA latent-world-model rollouts: same-state noised and unperturbed views should agree after the same action sequence, while action-distinct transitions remain separable.

Our use of ACPC is therefore narrower than an identifiability theorem, a learned bisimulation metric, or a robust MPC intervention. LeJEPA-style results motivate latent prediction and planning under latent recovery assumptions; bisimulation and value-aware models define task abstractions; group-action metrics test action faithfulness; ReOI modifies observation handling for robust visual MPC. This paper instead supplies a JEPA checkpoint diagnostic centered on additive Gaussian pixel-noise stress tests: predictive stability is tested on paired same-state clean/noised rollouts under the same action sequence, fixed-candidate cost stability is derived as a sufficient condition, and a separate discriminability guard checks that action-, transition-, or cost-distinct cases have not been collapsed.

3 Action-Conditioned Predictive Consistency

This section defines action-conditioned predictive consistency and its discriminability countercondition. We use ACPC to separate two requirements that encoder invariance alone conflates: same-state visual perturbations should agree after action-conditioned rollout, while task-grounded different-state pairs should remain separable. Section 3.2 then maps these two requirements to ATR and SMPR.

3.1 Setup and definitions

Let o_t be an unperturbed observation and $\tilde{o}_t = \tau(o_t)$ a visually perturbed view of the *same* underlying state (τ is additive Gaussian pixel noise here, but the definitions are perturbation-agnostic). Let h_t and $\tilde{h}_t$ be the corresponding observation–action histories, E_θ the encoder, and F_θ the action-conditioned latent predictor. Fix an action sequence $\mathbf{a}_t^H = a_{t:t+H-1}$ that is *identical* on both branches. For compact notation below, $\mathbf{a}_{0:k-1}$ denotes the length-k prefix of this sequence after re-indexing time at t, and we write

$$z_t = E_\theta(h_t), \quad \tilde{z}_t = E_\theta(\tilde{h}_t), \qquad \hat{z}_{t+k} = F_\theta^k(z_t, \mathbf{a}_{0:k-1}), \quad \hat{\tilde{z}}_{t+k} = F_\theta^k(\tilde{z}_t, \mathbf{a}_{0:k-1}). \tag{1}$$

Let Π denote a *task-relevant predictive projection* — the full latent, a transition delta, goal-distance or cost features, or a learned projection — and let d be a distance on its range.

Action-conditioned predictive consistency. For a same-state perturbation pair under the shared action sequence,

$$\mathrm{ACPC}_H(o_t, \tilde{o}_t, \mathbf{a}) = \sum_{k=1}^{H} \alpha_k\, d\big(\Pi(\hat{z}_{t+k}),\, \Pi(\hat{\tilde{z}}_{t+k})\big), \qquad \alpha_k \geq 0. \tag{2}$$

For the theoretical statements below, we use the induced horizon metric

$$d_H\big((u_1, \ldots, u_H), (v_1, \ldots, v_H)\big) = \sum_{k=1}^{H} \alpha_k d(u_k, v_k), \tag{3}$$

so that Equation (2) is exactly the clean/perturbed projected-rollout distance under the shared action sequence. In experiments, ATR uses the 90th percentile of this normalized same-state rollout disagreement. In this diagnostic framing, robustness corresponds to small ACPC_H together with action-relevant discriminability, not to the encoder distance $d(z_t, \tilde{z}_t)$ alone. Encoder closeness is a useful first-stage risk signal, but it is neither necessary (a large but task-irrelevant encoder shift can wash out through F_θ and Π) nor sufficient (a small encoder shift can still change the predicted future enough to flip the planned action).

Action-relevant discriminability (countercondition). Consistency must be *selective*. Let $Y^H(s, \mathbf{a})$ denote a task-relevant future signal that is external to the learned predictor — simulator state, cost-to-go, inverse-dynamics label, contact/keyframe tag, or another diagnostic proxy available in the dataset. For two genuinely different states s_i, s_j (with latents z_i, z_j) that admit an action sequence under which this external future signal diverges,

$$\Delta_{\mathrm{dyn}}^H(s_i, s_j, \mathbf{a}) = d_Y\big(Y^H(s_i, \mathbf{a}),\, Y^H(s_j, \mathbf{a})\big) > m, \tag{4}$$

the representation and predictions should keep them separable,

$$d\big(\Psi(z_i),\, \Psi(z_j)\big) > m', \tag{5}$$

where Ψ is a discriminability projection, possibly action-conditioned (latent, predicted delta under $\mathbf{a}$, inverse-dynamics feature, cost feature, or rollout embedding), and $m, m' > 0$ are margins. Equations (2)–(5) state the target: lower predictive disagreement for same-state visual perturbations must preserve discriminability among state/action/transition-distinct pairs. A method that drove $\mathrm{ACPC}_H \to 0$ by collapsing the latent would violate (5), so the two conditions are inseparable.

3.2 Operational consistency diagnostics

The reported diagnostic uses two metrics. ACPC Tail Risk (ATR) is the 90th percentile of normalized same-state clean/noisy ACPC-H rollout disagreement under a fixed recorded action sequence; lower is better. Selective Margin Pass Rate (SMPR) is the pass rate of task-grounded label-crossing pairs whose clean different-state rollout distance exceeds the same-state noisy radius at margin zero; higher is better.

The compressed diagnostic keeps only the two empirical quantities required by the selective-ACPC logic. ATR estimates the high-tail term $\Pr[D > \epsilon]$ that appears in the sampled-pool stability calibration, using the 90th percentile of normalized same-state action-conditioned rollout disagreement. SMPR estimates the discriminability countercondition by checking whether task-grounded near-boundary pairs remain separated beyond the same-state noisy radius. Thus low ATR without high SMPR is not interpreted as robustness, because collapse can also make clean and corrupted views agree. Encoder geometry and planner-side audits remain useful development checks, but they are not separate reported diagnostics in this compressed version.

3.3 ACPC as a fixed-candidate stability condition

ACPC is connected to planning through the candidate costs evaluated by model-predictive control. Let h be a clean history, $\tilde{h}$ the corresponding perturbed history, and let both branches evaluate the same candidate action sequence $\mathbf{a}$. For a goal or task descriptor g, define

$$C_h(\mathbf{a}, g) = J(\Pi(\hat{z}_{t+1}), \ldots, \Pi(\hat{z}_{t+H}), g)\,, \tag{6}$$

where J is the planner cost on the projected-rollout sequence. $C_{\tilde{h}}(\mathbf{a}, g)$ is defined analogously from the perturbed branch. The following statements are sufficient conditions for stability on a fixed candidate set; they do not prove stability of the full CEM sampling distribution or of the closed-loop trajectory over repeated replanning.

Proposition 1 (Encoder shift as one route to rollout disagreement)**.** *Fix an action sequence* $\mathbf{a}$ *and write*

$$G_{\mathbf{a}}(z) = \Pi(F_\theta^{1:H}(z, \mathbf{a}))$$

for the projected-rollout map. If $G_{\mathbf{a}}$ is locally L_G-Lipschitz between $E_\theta(h)$ and $E_\theta(\tilde{h})$ under encoder metric d_Z and projected-rollout metric d_H, then

$$d_H\big(G_{\mathbf{a}}(E_\theta(h)), G_{\mathbf{a}}(E_\theta(\tilde{h}))\big) \le L_G\, d_Z\big(E_\theta(h), E_\theta(\tilde{h})\big)\,.$$

Proof. This is the local Lipschitz condition applied to the two encoded histories. □

This bound makes encoder geometry a first-stage risk signal: small encoder shifts imply small rollout disagreement only under local insensitivity, large shifts can be contracted by $G_{\mathbf{a}}$, and small shifts can be amplified by the predictor or planner cost. This is why the empirical diagnostic measures the tail of action-conditioned rollout disagreement rather than encoder distance alone.

Proposition 2 (ACPC controls candidate-cost drift)**.** *Assume that J is locally L_J-Lipschitz on the clean/perturbed projected-rollout neighborhood evaluated by the fixed candidate set under metric d_H. If*

$$d_H\big(\Pi(F_\theta^{1:H}(E_\theta(h), \mathbf{a})), \Pi(F_\theta^{1:H}(E_\theta(\tilde{h}), \mathbf{a}))\big) \le \epsilon,$$

then

$$|C_h(\mathbf{a}, g) - C_{\tilde{h}}(\mathbf{a}, g)| \le L_J \epsilon.$$

Proof. The inequality is the local Lipschitz condition applied directly to the clean and perturbed projected-rollout sequences. □

In the compressed diagnostic, this condition is instantiated by the L2 rollout-token distance whose high-quantile tail defines ATR; empirical companion audits are kept outside the reported metric set.

Proposition 3 (Candidate top-1 stability)**.** *Let $\mathcal{A} = \{\mathbf{a}^1, \ldots, \mathbf{a}^K\}$ be the shared candidate set. Assume lower cost is preferred, and let $\mathbf{a}^{(1)}$ and $\mathbf{a}^{(2)}$ denote the lowest- and second-lowest-cost candidates on the clean branch. Suppose that for every candidate j,*

$$|C_h(\mathbf{a}^j, g) - C_{\tilde{h}}(\mathbf{a}^j, g)| \leq \eta.$$

If the clean branch has a strict top-1/top-2 margin

$$\Delta = C_h(\mathbf{a}^{(2)}, g) - C_h(\mathbf{a}^{(1)}, g) > 2\eta,$$

then the perturbed branch selects the same top-1 candidate.

Proof. For any candidate $\mathbf{a}^j \neq \mathbf{a}^{(1)}$,

$$C_{\tilde{h}}(\mathbf{a}^j, g) - C_{\tilde{h}}(\mathbf{a}^{(1)}, g) \geq C_h(\mathbf{a}^j, g) - C_h(\mathbf{a}^{(1)}, g) - 2\eta \geq \Delta - 2\eta > 0.$$

Thus every non-top candidate remains more costly than the clean top-1 candidate on the perturbed branch. □

Corollary 1 (ACPC-margin condition for candidate stability)**.** *Let $\mathcal{A} = \{\mathbf{a}^1, \ldots, \mathbf{a}^K\}$ be a shared candidate set and assume the planner cost J is locally L_J-Lipschitz as in Proposition 2. If every candidate in $\mathcal{A}$ has projected-rollout discrepancy at most ϵ between the clean and perturbed branches, and the clean branch has top-1/top-2 margin $\Delta > 2L_J\epsilon$, then the two branches select the same top-1 candidate from $\mathcal{A}$.*

Proof. Proposition 2 gives $|C_h(\mathbf{a}^j, g) - C_{\tilde{h}}(\mathbf{a}^j, g)| \leq L_J\epsilon$ for every candidate. Applying Proposition 3 with $\eta = L_J\epsilon$ gives the result. □

The corollary gives a formal link between ACPC and fixed-candidate decisions without turning the diagnostic into a closed-loop guarantee. It connects paired predictive agreement to candidate-cost stability and then to action selection under a margin condition. The result is limited to a fixed candidate set; adaptive sampling, repeated replanning, and environment feedback can still change the closed-loop trajectory.

3.4 Sampled-pool stability and Gaussian sensitivity

The fixed-candidate statement is the deterministic core. A sampled candidate pool adds finite-sample risk. The next bound keeps the same diagnostic scope while exposing why ATR must be a tail metric and why clean candidate margins remain a caveat.

Theorem 1 (Sampled-pool ACPC stability)**.** *Let $\mathcal{A} = \{\mathbf{a}^1, \ldots, \mathbf{a}^K\} \sim q^K$ be a once-sampled candidate pool, and use deterministic tie-breaking for* $\arg\min$*. For each sampled candidate define*

$$D_j = d_H\left(\Pi(F_\theta^{1:H}(E_\theta(h), \mathbf{a}^j)), \Pi(F_\theta^{1:H}(E_\theta(\tilde{h}), \mathbf{a}^j))\right).$$

Assume J is locally L_J-Lipschitz on the evaluated projected-rollout neighborhood and that a single draw from q satisfies

$$\Pr_{\mathbf{a}\sim q}[D(\mathbf{a}) > \epsilon] \leq \delta.$$

Let $\Delta_{\mathcal{A}} = C_h(\mathbf{a}^{(2)}, g) - C_h(\mathbf{a}^{(1)}, g)$ be the clean branch top-1/top-2 margin inside the sampled pool. Then

$$\Pr_{\mathcal{A}}\left[\arg\min_j C_h(\mathbf{a}^j, g) \neq \arg\min_j C_{\tilde{h}}(\mathbf{a}^j, g)\right] \leq K\delta + \Pr_{\mathcal{A}}[\Delta_{\mathcal{A}} \leq 2L_J\epsilon].$$

Theorem 1 is not a convergence theorem for adaptive planning. It controls the instability of one sampled candidate pool. The first term is the probability that at least one candidate has a large clean/noisy rollout discrepancy, which motivates ATR. The second term is the probability that the clean candidate pool has too small a top-1 margin; it explains why ATR is not a closed-loop guarantee. SMPR is separate from this flip bound: it is the selective-discriminability condition required to make low ATR meaningful rather than collapsed.

Corollary 2 (Replanning union bound)**.** *If replanning step t has sampled-pool flip-probability bound*

$$r_t = K_t\delta_t + \Pr[\Delta_{\mathcal{A}_t} \leq 2L_J\epsilon_t],$$

then over T replanning steps,

$$\Pr[\exists t \leq T : \textit{ the sampled-pool top-1 flips at step } t] \leq \sum_{t=1}^{T} r_t.$$

Proposition 4 (Finite-sample tail calibration for paired diagnostics)**.** *Fix the diagnostic sampling distribution used to draw independent paired clean/noisy histories and candidate action sequences. Let*

$$X_i = \mathbf{1}[D_i > \epsilon], \qquad i = 1, \ldots, n,$$

where D_i is the paired ACPC projected-rollout discrepancy for diagnostic item i. Let $p_\epsilon = \Pr[D > \epsilon]$ and $\hat{p}_\epsilon = n^{-1}\sum_i X_i$. Then, with probability at least $1-\delta$ over the diagnostic sample,

$$p_\epsilon \leq \hat{p}_\epsilon + \sqrt{\frac{\log(1/\delta)}{2n}}.$$

The same one-sided calibration applies to SMPR after fixing its task-grounded pair construction.

This calibration is with respect to the fixed diagnostic sampling distribution and its independent-sample idealization. It does not convert an empirical quantile or pass-rate table into a uniform bound over all closed-loop states; when nearby dataset windows are dependent, we use it only as calibration.

The Gaussian stressor also has a local sensitivity interpretation. Let $G_{\mathbf{a}}(z) = \Pi(F_\theta^{1:H}(z, \mathbf{a}))$ and let $\tilde{o} = o + \xi$ with $\xi \sim \mathcal{N}(0, \sigma^2 I)$.

Proposition 5 (Local Gaussian ACPC sensitivity)**.** *If E_θ and $G_{\mathbf{a}}$ are locally twice differentiable around o and $E_\theta(o)$ with bounded second-order remainders, then for small σ,*

$$G_{\mathbf{a}}(E_\theta(o+\xi)) - G_{\mathbf{a}}(E_\theta(o)) = J_{G_{\mathbf{a}}}(E_\theta(o))J_E(o)\xi + R_\xi,$$

where the remainder contributes higher-order terms, and

$$\mathbb{E}_\xi\left[\|G_{\mathbf{a}}(E_\theta(o+\xi)) - G_{\mathbf{a}}(E_\theta(o))\|_2^2\right] = \sigma^2\, \|J_{G_{\mathbf{a}}}(E_\theta(o))J_E(o)\|_F^2 + O(\sigma^3).$$

Thus Gaussian ACPC measures action-conditioned encoder–predictor sensitivity, not encoder invariance alone. Small encoder shifts can still be amplified by $J_{G_{\mathbf{a}}}$, while large nuisance shifts can become harmless after rollout if the product $J_{G_{\mathbf{a}}}J_E$ is small in those directions. The selective target is therefore asymmetric: the product should be small along nuisance perturbation directions, while projected rollouts for action-relevant state differences remain separated. The task-grounded near-boundary proxy margin pass event used later is

$$M = \mathbf{1}[d_{\text{proxy diff}} > d_{\text{same state noise}} + \delta_m]\,,$$

with $\delta_m = 0$ in the reported table. It is a finite state-proxy selective-consistency check, not a full semantic-label coverage guarantee.

The theoretical statements in this section are intended as diagnostic calibration rather than as closed-loop control guarantees. They identify the failure modes that a paired clean/noisy diagnostic should measure: rollout-disagreement tails, clean candidate-margin caveats, and collapse of task-grounded separations. They do not assert that a low empirical ATR is a uniform bound for adaptive planning, repeated replanning, or environment-feedback stability.

Selective ACPC pseudo-metric. For a nonempty fixed candidate family $\mathcal{A}$, the stability condition can be summarized by

$$D^{\mathcal{A}}_{\text{ACPC}}(h, \tilde{h}) = \max_{\mathbf{a}\in\mathcal{A}} d_H\left(\Pi(F_\theta^{1:H}(E_\theta(h), \mathbf{a})), \Pi(F_\theta^{1:H}(E_\theta(\tilde{h}), \mathbf{a}))\right). \tag{7}$$

When d_H is a metric (or a pseudo-metric, if some horizon weights vanish), $D^{\mathcal{A}}_{\text{ACPC}}$ is a pseudo-metric on histories after passing through the encoder, predictor, and projection: distinct histories can have zero distance if all candidate rollouts are projection-equivalent. This follows because each map $h \mapsto \Pi(F_\theta^{1:H}(E_\theta(h), \mathbf{a}))$ pulls back d_H to a pseudo-metric, and the pointwise maximum of pseudo-metrics is again a pseudo-metric. The useful object is therefore *selective* predictive stability. Small $D^{\mathcal{A}}_{\text{ACPC}}$ for same-state visual perturbations gives candidate-cost and top-1 stability under the preceding Lipschitz and margin assumptions; the same contraction is not desirable for state/action pairs whose external task variables differ, which is why the discriminability guard is part of the definition rather than an auxiliary aesthetic check.

Corollary 3 (Fixed-candidate predictive stability)**.** *Assume $D^{\mathcal{A}}_{\text{ACPC}}(h, \tilde{h}) \leq \epsilon$ for a same-state visual-perturbation pair and that J is locally L_J-Lipschitz on the projected rollout. If the clean candidate margin satisfies $\Delta > 2L_J\epsilon$, then the clean and perturbed branches select the same top-1 candidate from the fixed set $\mathcal{A}$.*

Proof. By definition of $D^{\mathcal{A}}_{\mathrm{ACPC}}$, every candidate in $\mathcal{A}$ has projected-rollout discrepancy at most ϵ. Proposition 2 bounds the cost drift of every candidate by $L_J\epsilon$, and Proposition 3 gives top-1 stability when $\Delta > 2L_J\epsilon$. ☐

The statement becomes selective only when it is paired with a discriminability condition such as Equation (5). Without that second condition, fixed-candidate predictive stability can still be achieved by a collapsed encoder–predictor, as shown next.

Encoder closeness is not the right substitute for this condition. It is not necessary: a nuisance-subspace change can move $E_\theta(h)$ and $E_\theta(\tilde{h})$ far apart while leaving $\Pi(F_\theta^{1:H}(\cdot, \mathbf{a}))$ and the cost unchanged. It is not sufficient either: a small encoder displacement can flip a low-margin cost ranking when the predictor or planner cost has high local sensitivity.

Proposition 6 (ACPC alone permits collapse)**.** *There exist encoder–predictor pairs for which* $\mathrm{ACPC}_H = 0$ *for every same-state perturbation pair and every action sequence, while action-distinct states are mapped to identical projected rollouts.*

Proof. Let $E_\theta(h) = z_0$ for every history and let $F_\theta^k(z_0, \mathbf{a}) = u_0$ for every horizon k and action sequence $\mathbf{a}$. Then clean and perturbed branches always have identical projected rollouts, so same-state ACPC is zero. However, any two states that require different actions, costs, or transitions also have identical projected rollouts, violating the discriminability condition in Equation (5). ☐

Thus low ACPC is meaningful only with a guard on action-relevant separation. This is the role of SMPR in the reported diagnostic. This is where ACPC differs from existing nuisance-feature analyses of latent prediction: those results help explain why latent prediction may suppress irrelevant or noisy features at the representation level, whereas the condition above asks whether the remaining perturbation changes action-conditioned projected rollouts while preserving task-grounded separations.

4 Study protocol

The protocol separates behavior endpoints from diagnostics. Closed-loop success under Gaussian evaluation noise establishes the behavioral phenomenon; ATR and SMPR are then computed on fixed checkpoints to localize whether recovered endpoints also satisfy the selective ACPC pattern. Training seed, checkpoint epoch, evaluation seeds, and trajectory counts are kept fixed across the main comparisons.

4.1 Models and noise intervention

LeWM [19] is an end-to-end JEPA world model trained with a latent prediction loss and SIGReg regularization for latent-space model-predictive control. PushT (2D pushing), TwoRoom (2D navigation), Reacher (2D arm), and Cube (3D manipulation) form a heterogeneous stress panel spanning different nuisance/discriminability regimes. We add per-frame Gaussian noise to the input sequence as a full-sequence augmentation: each frame is independently noised with probability $p = 1.0$, and the standard deviation is drawn from $\mathrm{Uniform}(0, \sigma_{\max})$. This is a full-sequence input-side augmentation. We sweep $\sigma_{\max} \in \{0.01, \ldots, 0.08\}$ and evaluate observation-only Gaussian noise with the goal image unperturbed as the primary robustness endpoint. Goal-corrupted Gaussian evaluation is retained only as an auxiliary appendix stress condition.

4.2 Evaluation and diagnostics

The canonical LeWM development grid contains 36 epoch-10 checkpoints for training seed 3072: four tasks and, for each task, a no-noise baseline plus eight noise-trained checkpoints. Each checkpoint is evaluated with three evaluation seeds (42/43/44) and 100 trajectories per seed; canonical tables report the population mean ± population std across those three evaluation seeds. A completed Gaussian training-seed replication adds independent LeWM training seeds 3073 and 3074 under the same nine-point Gaussian sweep and evaluation protocol. Together, seed 3072 plus replication seeds 3073/3074 provide three training seeds for the main Gaussian closed-loop endpoint in Table 2.

The main results follow the compressed evidence chain used in the claims. Closed-loop tables report training-seed mean/std for Gaussian robustness. ATR measures whether same-state clean/noisy rollout-disagreement tails contract at recovered endpoints, and SMPR checks whether task-grounded near-boundary separations survive that contraction. Unseen-stressor scores are auxiliary scope checks rather than additional core diagnostics.

5 Experiments

The experiments use additive Gaussian pixel noise as the primary controlled stressor for a representation-diagnostic question. We first show that the stressor changes closed-loop behavior. We then ask where recovery appears in the model, prioritizing cross-task ACPC/paired-rollout evidence and keeping task-specific representation measurements only as discriminability checks. The study is a matched-stressor trace through the encoder–predictor–planner chain; point-best noise-level ranking, external method comparison, and broad perturbation-family transfer are outside its scope.

All results use the protocol in Section 4: LeWM-base denotes the no-noise checkpoint, and LeWM+noise denotes the eight-level $\sigma_{\max}$ sweep.

5.1 JEPA control fragility under Gaussian observation noise

Table 1 reports LeWM-base success rates under observation-only Gaussian evaluation noise. Each training-seed value is first averaged over the three evaluation seeds; the table reports mean $\pm$ population std across the three training seeds. The goal image is kept clean in every column.

Table 1: LeWM-base under observation-only Gaussian evaluation noise. Values are mean $\pm$ population std across training seeds 3072/3073/3074; each training-seed value averages evaluation seeds 42/43/44 with 100 trajectories per evaluation seed. The goal image is kept clean.

Task	eval $\sigma = 0$	eval $\sigma = 0.03$	eval $\sigma = 0.05$	eval $\sigma = 0.08$
TwoRoom	94.67 ± 0.72	92.56 ± 1.23	81.33 ± 6.77	68.78 ± 2.45
PushT	81.78 ± 4.89	53.56 ± 12.91	19.22 ± 9.12	7.22 ± 5.81
Reacher	59.22 ± 1.03	42.56 ± 2.99	33.56 ± 1.77	18.22 ± 0.42
Cube	65.22 ± 1.10	63.00 ± 1.19	46.67 ± 1.96	43.11 ± 3.27

LeWM-base is strong on unperturbed images, but observation noise alone produces large closed-loop losses at $\sigma = 0.08$: PushT loses 74.56 ± 5.55 points, Reacher 41.00 ± 1.44 points, TwoRoom 25.89 ± 2.38 points, and Cube 22.11 ± 2.78 points between the first and last columns. The progression across noise levels is smooth enough to show that the endpoint is not an isolated artifact of one severity setting. PushT degrades most sharply, consistent with contact-heavy continuous control being sensitive to visual precision.

5.2 Noise augmentation supplies recovered checkpoints for diagnosis

Figure 1 shows the full seed-3072 Gaussian sweep. The main qualitative pattern is a broad, task-dependent recovery plateau rather than a stable point-best training-noise level. Exact point-best rows are artifact-level bookkeeping rather than a ranking claim.

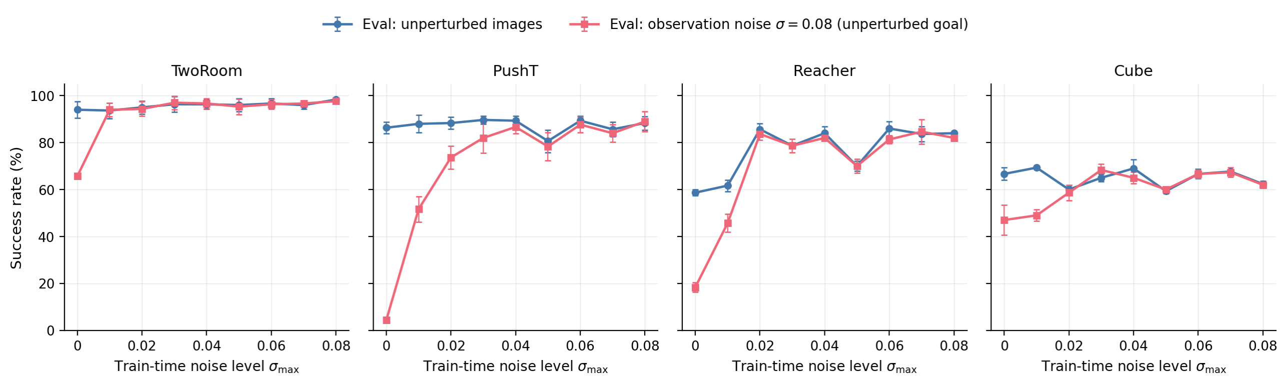

Figure 1: Noise-training sweep across four tasks. Blue circles: unperturbed evaluation; red squares: observation-only Gaussian noise $\sigma = 0.08$ with an unperturbed goal. Error bars show the population standard deviation across the three evaluation seeds. The curves document broad task-dependent recovery plateaus rather than point-optimal ranking.

The sweep contributes two observations to the main argument. First, input-side noise can be an effective intervention in this protocol: PushT recovers from 4.33% to 89.00% at observation-noise 0.08, and Reacher from

18.33% to 84.67%. Second, the grid supplies representative recovered checkpoints for the downstream diagnostic question: what changed in the encoder–predictor–planner chain?

Reacher also shows a fixed-epoch clean-control lift. Noise training improves not only observation-noise robustness but also unperturbed control: the representative clean row rises from 58.67% at the no-noise baseline to about 86% under noise training. We interpret this as a fixed-protocol regularization or stabilization effect rather than pure robustness recovery. The three-training-seed replication below makes the endpoint less dependent on seed 3072, but longer learning curves would still be needed to separate regularization from optimization timing.

The completed Gaussian training-seed replication strengthens the closed-loop part of the claim. It repeats the full Gaussian sweep for LeWM seeds 3073 and 3074 and combines them with the canonical seed-3072 sweep. Table 2 summarizes the observation-only Gaussian endpoint across all three training seeds.

Table 2: Three-training-seed Gaussian endpoint for LeWM. Values are mean $\pm$ population std across training seeds 3072/3073/3074; each seed point averages evaluation seeds 42/43/44 with 100 trajectories per evaluation seed. The endpoint is observation-only Gaussian evaluation at $\sigma = 0.08$ with a clean goal image.

Task	base obs $\sigma = 0.08$	std0.08 obs $\sigma = 0.08$	gain	best obs $\sigma = 0.08$	std0.08 gap
TwoRoom	68.78 ± 2.46	97.11 ± 0.57	28.33 ± 2.68	97.67 ± 0.27	0.56 ± 0.79
PushT	7.22 ± 5.81	85.78 ± 3.45	78.56 ± 5.18	87.67 ± 0.98	1.89 ± 2.67
Reacher	18.22 ± 0.42	81.55 ± 0.88	63.34 ± 1.25	83.78 ± 0.63	2.22 ± 0.87
Cube	43.11 ± 3.27	62.56 ± 1.55	19.45 ± 4.53	66.44 ± 1.34	3.89 ± 2.20

Across three training seeds, the no-noise checkpoints retain large observation-noise cliffs, while the fixed $\sigma_{\max} = 0.08$ endpoint is within 0.56–3.89 pp of each task's best observed obs-$\sigma = 0.08$ row. The task-level best rows occur within broad ranges: 0.05–0.08 for TwoRoom, 0.04–0.08 for PushT, 0.05–0.07 for Reacher, and 0.03–0.05 for Cube. PushT remains the sharpest stress case, improving by 78.56 ± 5.18 points at the fixed high-noise endpoint; Reacher improves by 63.34 ± 1.25 points. These gaps and ranges support plateau wording rather than a universal optimum at exactly $\sigma_{\max} = 0.08$.

A three-seed blur/resize score check later in this section serves as an auxiliary severe-stressor check outside the matched-Gaussian endpoint. The next subsection asks whether the recovered Gaussian endpoints also move in the two compressed diagnostics.

5.3 Compressed selective-ACPC diagnostics

For the main diagnostic, ATR is the 90th percentile of same-state clean/noisy ACPC-H rollout distance normalized by the clean transition scale. SMPR is the task-grounded near-boundary selective-margin pass rate at margin zero. Both are computed without retraining on the fixed no-noise and $\sigma_{\max} = 0.08$ checkpoints for training seeds 3072/3073/3074. Table 3 reports population mean/std across training seeds.

Table 3: Compressed selective-ACPC diagnostics over training seeds 3072/3073/3074. ATR is the q90 normalized same-state clean/noisy action-conditioned rollout disagreement; lower is better. SMPR is the task-grounded near-boundary selective-margin pass rate; higher is better.

Task	ATR base	ATR std0.08	SMPR base	SMPR std0.08
TwoRoom	1.509 ± 0.010	0.111 ± 0.016	0.34 ± 0.05	0.99 ± 0.01
PushT	3.580 ± 0.260	0.247 ± 0.017	0.44 ± 0.17	1.00 ± 0.00
Reacher	2.628 ± 0.048	0.082 ± 0.005	0.73 ± 0.03	1.00 ± 0.00
Cube	2.320 ± 0.045	0.100 ± 0.007	0.45 ± 0.03	1.00 ± 0.00

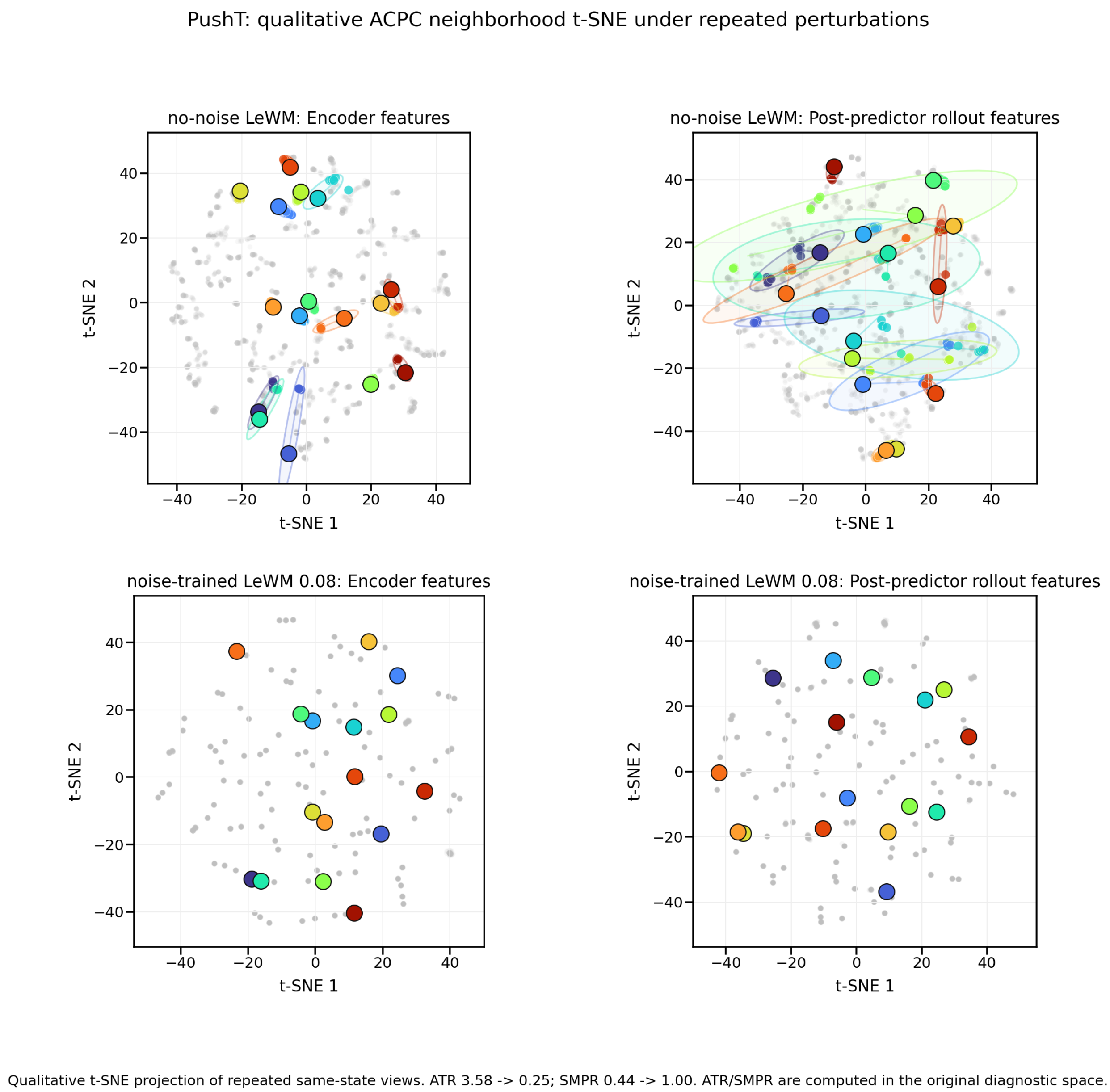

Figure 2: Qualitative PushT ACPC neighborhood t-SNE visualization for the no-noise versus $\sigma_{\max} = 0.08$ contrast. The four panels show encoder features and post-predictor rollout features for repeated same-state perturbation views; colored clusters mark fixed random anchors. The t-SNE projection is qualitative. ATR and SMPR are the quantitative diagnostics and are computed in the original projected-rollout diagnostic space.

The two diagnostics move in the intended selective direction on all four tasks: recovered endpoints have much smaller same-state predictive tails and much higher task-grounded margin pass rates. Figure 2 provides qualitative visual intuition on PushT: repeated noisy views of the same state become tighter after noise training in both encoder and post-predictor feature projections. The quantitative claim still rests on ATR, SMPR, and closed-loop behavior, not on distances in the t-SNE plane. Cube remains the useful boundary case: the compressed diagnostics improve strongly, but the closed-loop Gaussian gain and gap in Table 2 are weaker than PushT or Reacher, so closed-loop evaluation remains the behavioral authority.

SMPR uses programmatic task-state proxy labels, not oracle semantic annotations: TwoRoom uses doorway/room-side labels, PushT uses T-block pose cells, Reacher uses target/end-effector relation bins, and Cube uses cube-pose/goal-relation cells. Each anchor selects the closest label-crossing neighbor inside the closest 35% state-distance neighborhood and passes if that clean different-state rollout distance exceeds the same-state noisy radius. The no-noise baselines are weak under this stricter audit, while the high-noise endpoint reaches near-one pass rate; stronger hand-labeled or simulator-derived contact, topology, action-value, or cost-to-go labels remain future validation.

5.4 Bounded unseen-stressor check

The main evidence remains the matched Gaussian-noise endpoint. After freezing the Gaussian grid, we ran strongest-severity unseen-stressor score checks to test whether the fixed $\sigma_{\max} = 0.08$ endpoint transfers outside the training perturbation family. Table 4 reports behavior only. TwoRoom and Reacher improve under strongest blur, whereas PushT and Cube do not show a clear resize gain at the scale of training-seed and evaluation variance. We treat this as bounded behavior outside the matched Gaussian setting.

Table 4: Bounded unseen-stressor score check. Values are mean $\pm$ population std across LeWM training seeds 3072/3073/3074; each seed point averages evaluation seeds 42/43/44 from strongest-severity unseen runs. The two score columns compare the no-noise LeWM checkpoint with the fixed $\sigma_{\max} = 0.08$ noise-trained checkpoint under the same non-Gaussian evaluation stressor.

Task	evaluation stressor	no-noise ckpt	$\sigma_{\max} = 0.08$ ckpt
TwoRoom	blur $k = 15$	47.67 ± 5.44	90.78 ± 5.38
Reacher	blur $k = 15$	22.00 ± 3.78	71.22 ± 1.10
PushT	resize 0.25	63.44 ± 14.05	66.33 ± 8.38
Cube	resize 0.25	57.00 ± 1.96	56.11 ± 0.57

The pattern is mixed outside the training perturbation family: the $\sigma_{\max} = 0.08$ checkpoint gives large gains for TwoRoom and Reacher under blur, while PushT and Cube are essentially unchanged under the resize stressor. These rows therefore delimit the Gaussian result rather than extending it into a general perturbation-transfer claim.

6 Discussion and limitations

What the evidence supports. This is a controlled diagnostic study, not a method paper. Closed-loop evaluation establishes the behavior; ATR checks whether same-state noisy views have small action-conditioned predictive tails; SMPR checks whether task-grounded near-boundary distinctions survive that contraction. The primary statistic is the full three-training-seed Gaussian replication, with evaluation-seed variance kept as the within-seed measurement scale. These standard deviations are descriptive over three training seeds, not asymptotic significance claims.

Scope. Gaussian pixel noise remains the training/evaluation axis. The blur/resize rows are bounded behavior outside the matched Gaussian setting, not a transfer theorem. ATR/SMPR are not a point-best $\sigma_{\max}$ predictor and do not replace closed-loop evaluation inside a broad plateau. We do not claim superiority over DrQ-style augmentation, TD-MPC2, DreamerV3, or robust MPC methods; the contribution is a no-retraining diagnostic for JEPA latent world-model checkpoints under a controlled Gaussian stressor.

What remains. The results argue against a too-strong reading of latent prediction: future-latent prediction alone does not guarantee closed-loop visual robustness. Stronger discriminability evidence would require hand-labeled or simulator-derived contact, topology, action-value, or cost-to-go labels, especially for PushT and Cube. Future methods can turn ATR/SMPR into objectives, but this paper keeps that as a separate method-paper path after diagnostic validity is established.

7 Conclusion

This paper frames Gaussian-noise robustness in JEPA latent world-model control as a selective diagnostic problem. Closed-loop behavior establishes the phenomenon; ATR measures same-state predictive-stability tails; SMPR checks task-grounded anti-collapse margins. On fixed LeWM checkpoints, input-side noise training recovers broad matched-Gaussian robustness plateaus across three training seeds, and the two compressed diagnostics move in the expected direction at recovered endpoints.

The theory is deliberately diagnostic: sampled-pool ACPC exposes why tail risk matters and why clean margins prevent closed-loop guarantees, while local Gaussian linearization connects the measured tail to action-conditioned encoder–predictor sensitivity. The broader implication is not solved generalization; it is a concrete audit target for future methods: stable action-conditioned predictive dynamics under nuisance variation, while preserving distinctions needed for planning.

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A Proofs and Calibration for ACPC Diagnostics

Reading. These proofs calibrate ATR and SMPR as diagnostics; they do not provide closed-loop control guarantees.

Proof of Proposition 2. For any fixed action sequence $\mathbf{a}$, the assumption gives $d_H(U_h(\mathbf{a}), U_{\tilde{h}}(\mathbf{a})) \leq \epsilon$. The planner cost J is L_J-Lipschitz on the projected-rollout sequence, so $|C_h(\mathbf{a}, g) - C_{\tilde{h}}(\mathbf{a}, g)| \leq L_J\epsilon$. Taking the maximum over the fixed candidate set gives the claim. □

Proof of Proposition 3. Let $\mathbf{a}^\star$ be the clean best candidate and $\mathbf{a}^{(2)}$ the clean runner-up. If every candidate cost moves by at most $L_J\epsilon$ and the clean margin is larger than $2L_J\epsilon$, then the perturbed cost of $\mathbf{a}^\star$ remains below the perturbed cost of every other candidate. Thus the top candidate is unchanged. □

Proof of Theorem 1. For a sampled pool of size K, the probability that at least one candidate has clean/noisy rollout discrepancy above ϵ is at most $K\delta$ by the union bound. On the complement of that event, every candidate cost moves by at most $L_J\epsilon$. A flip can then occur only if the clean top margin is at most $2L_J\epsilon$, which contributes the margin-tail term. This proves the stated sampled-pool instability bound. □

Finite-sample calibration. For a bounded Bernoulli diagnostic with empirical pass rate $\hat{p}$ over n fixed pairs, Hoeffding gives

$$\Pr\left[p < \hat{p} - \sqrt{\frac{\log(1/\alpha)}{2n}}\right] \leq \alpha. \tag{8}$$

We use this only as calibration for the fixed SMPR pair construction; nearby dataset windows can be dependent.

Local Gaussian sensitivity. For small Gaussian pixel noise ξ, a first-order expansion gives $E_\theta(o + \xi) = E_\theta(o) + J_E(o)\xi + \mathcal{R}(\xi)$ with $\|\mathcal{R}(\xi)\| = O(\|\xi\|^2)$. Composing the encoder with the action-conditioned rollout map shows that same-state rollout disagreement measures local sensitivity of the encoder–predictor composition, not encoder invariance alone. ATR reports a high-quantile version of that paired rollout disagreement.

B Experimental Details

Reading. These details define the fixed evaluation and diagnostic protocol used by the main tables.

LeWM is evaluated on PushT, TwoRoom, Reacher, and Cube using the canonical three evaluation seeds 42/43/44 and 100 trajectories per seed. The main Gaussian robustness endpoint perturbs observation pixels with Gaussian noise at standard deviation 0.08 while keeping the goal image clean. Noise-trained checkpoints use full-sequence input-side Gaussian augmentation with $\sigma_{\max} \in \{0.01, \ldots, 0.08\}$.

ATR is computed at each task–training-seed–checkpoint row as the 90th percentile same-state ACPC-H distance normalized by the clean transition scale. SMPR uses task-grounded near-boundary proxy labels at margin zero. The paper reports population mean/std across training seeds 3072/3073/3074.

C Auxiliary Observation+Goal Gaussian Stress

Reading. This auxiliary table reports a harder Gaussian stress condition where the goal image is also perturbed. It explains why the main endpoint keeps the goal clean and does not broaden the main matched-observation claim.

Table 5: LeWM-base under auxiliary observation+goal Gaussian evaluation noise at $\sigma = 0.08$. Values are mean $\pm$ population std across training seeds 3072/3073/3074; each training-seed value averages evaluation seeds 42/43/44 with 100 trajectories per evaluation seed.

Task	observation-only $\sigma = 0.08$	observation+goal $\sigma = 0.08$
TwoRoom	68.78 ± 2.45	53.56 ± 2.57
PushT	7.22 ± 5.81	4.89 ± 1.64
Reacher	18.22 ± 0.42	13.78 ± 1.50
Cube	43.11 ± 3.27	45.78 ± 0.79

Perturbing both streams is generally harsher for TwoRoom, PushT, and Reacher, while Cube is within the observed training-seed variation. The main experiments keep the goal image clean so that the endpoint isolates robustness to corrupted observations rather than changing both the current observation and the target descriptor.